MATH 115 Sample Midterm 2

Questions 1 - 6 are multiple choice questions. For each of these questions, there is exactly one correct answer.

1. Consider the system of linear equations

$$\begin{array}{rcrcrcl} 7x_1 & + & 3x_2 & - & x_3 & = & 4 \\ 5x_1 & - & x_2 & + & 4x_3 & = & 6 \end{array}.$$

Which of the following are solutions to this system?

(a) $\vec{x} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 3 \\ 2 \end{bmatrix}, \quad t \in \mathbb{R}.$

(b) $\vec{x} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ -3 \\ -2 \end{bmatrix}, \quad t \in \mathbb{R}.$

(c) $\vec{x} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -2 \\ 6 \\ 4 \end{bmatrix}, \quad t \in \mathbb{R}.$

(d) All of the above.

(e) None of the above.

2. Let $A \in M_{m\times n}(\mathbb{R})$, $\vec{b} \in \mathbb{R}^m$ and let R be the reduced row echelon form of A. Consider the system of linear equations $A\vec{x} = \vec{b}$. Which of the following must be true?

(a) If the system $A\vec{x} = \vec{b}$ has infinitely many solutions, then R has a row of zeros.

(b) If R has a row of zeros, then $A\vec{x} = \vec{b}$ has infinitely many solutions.

(c) If $A\vec{x} = \vec{b}$ is consistent, then $\vec{b}$ can be expressed as a linear combination of the columns of A.

(d) If $A\vec{x} = \vec{b}$ has a unique solution, then $m = n$.

(e) None of the above.

3. If $A\vec{x} = \vec{b}$ is inconsistent where A is a square matrix, then the system $A\vec{x} = \vec{0}$ has

(a) no solutions.

(b) a single solution.

(c) infinitely many solutions.

(d) exactly rank(A) solutions.

(e) None of the above.

4. Let $A = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}$. Which of the following is true?

   (a) $A^{-1} = \begin{bmatrix} 4 & -2 \\ -2 & 4 \end{bmatrix}$.

   (b) $A^{-1} = \begin{bmatrix} 1/3 & -1/6 \\ -1/6 & 1/3 \end{bmatrix}$.

   (c) $A^{-1} = \begin{bmatrix} 2 & 4 \\ 4 & 2 \end{bmatrix}$.

   (d) $A^{-1} = \begin{bmatrix} 1 & -1/2 \\ -1/2 & 1 \end{bmatrix}$.

   (e) None of the above.

5. Let $\left[A \mid \vec{b} \right]$ be the augmented matrix of a system of m linear equations in n variables. Consider the following three statements.

   I) The system is consistent if and only if $\text{rank}(A) = \text{rank}\left([\, A \,|\, \vec{b}\,]\right)$.

   II) If the system is consistent, then the general solution to this system will have $n - \text{rank}(A)$ parameters.

   III) If $\text{rank}(A) = m$, then the system is consistent.

   Choose the correct answer from the choices below.

   (a) I) and II) are true, but III) is false.

   (b) II) and III) are true, but I) is false.

   (c) II) is true, but I) and III) are false.

   (d) I), II), and III) are all true.

   (e) I), II), and III) are all false.

6. Let $A \in M_{2\times 2}(\mathbb{R})$ be such that $\text{rank}(A) = 1$. Which of the following is **false**?

   (a) $\text{rank}(A^2) = 1$.

   (b) If the system $A\vec{x} = \vec{b}$ is consistent, then it will have infinitely many solutions

   (c) The system $A\vec{x} = \vec{b}$ will be inconsistent for some choice of $\vec{b} \in \mathbb{R}^2$.

   (d) Any REF of A has a row of zeros.

   (e) None of the above.

Questions 7 - 12 are long answer questions.

7. Let
$$\vec{u} = \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix} \quad \text{and} \quad \vec{w} = \begin{bmatrix} -2 \\ -4 \\ 1 \end{bmatrix}.$$
   (a) Compute $4\vec{u} - 2\vec{w}$.
   (b) Compute the normalization of $\vec{u}$, that is, compute $\widehat{u}$.
   (c) Determine if $\vec{u}$ and $\vec{w}$ determine an acute angle, an obtuse angle, or if they are orthogonal.
   (d) Find a scalar equation for the plane given by
$$\vec{x} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} + s\vec{u} + t\vec{w}, \quad s, t \in \mathbb{R}.$$

8. Consider the line in $\mathbb{R}^3$ with vector equation
$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ -4 \end{bmatrix} + t \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \quad t \in \mathbb{R}.$$
   (a) Find the shortest distance from the point $P(1, -5, 6)$ to the line L.
   (b) Find the point Q on L that is closest to P.
   (c) Find a vector equation of the plane T that contains both the line L and the point P.

9. Consider the system
$$\begin{array}{rcrcrcr} x_1 & + & 2x_2 & + & \alpha x_3 & = & 3 \\ 2x_1 & + & x_2 & + & x_3 & = & \beta \\ -x_1 & + & x_2 & + & 2x_3 & = & 1 \end{array}$$
   Determine, with justification, the values of the constants $\alpha, \beta \in \mathbb{R}$ so that the system has
   (a) no solutions.
   (b) a unique solution.
   (c) infinitely many solutions.

10. Consider the linear system
$$\begin{array}{rcrcrcrcr} x_1 & - & 3x_2 & - & 3x_3 & + & x_4 & = & 1 \\ 2x_1 & - & 4x_2 & - & 2x_3 & + & 4x_4 & = & -2 \end{array}$$
    (a) Solve this system by carrying the augmented matrix to RREF. What does the solution set represent geometrically?
    (b) Write the general solution of the associated homogeneous system.
    (c) State the rank of the coefficient matrix of the system.

11. (a) Let

$$\vec{a}_1 = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} \quad \text{and} \quad \vec{a}_2 = \begin{bmatrix} -1 \\ 4 \\ 1 \end{bmatrix}.$$

Find $\vec{a}_3 \in \mathbb{R}^3$ given that $A = \begin{bmatrix} \vec{a}_1 & \vec{a}_2 & \vec{a}_3 \end{bmatrix}$ and $A \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$.

(b) Find the 2×2 matrix B if

$$\left(\begin{bmatrix} 2 & -1 \\ 3 & -4 \end{bmatrix} + (2B)^T \right)^{-1} = \begin{bmatrix} 2 & 9 \\ 1 & 4 \end{bmatrix}.$$

You must use the Matrix Inversion Algorithm if you compute any inverses.

12. (a) Let $\vec{x}, \vec{y} \in \mathbb{R}^n$. Prove that $\|\vec{x} + \vec{y}\|^2 + \|\vec{x} - \vec{y}\|^2 = 2\left(\|\vec{x}\|^2 + \|\vec{y}\|^2\right)$

(b) Let $\vec{v}_1, \vec{v}_2 \in \mathbb{R}^n$. Prove that $\vec{x}$ can be written as a linear combination of $\vec{v}_1$ and $\vec{v}_2$ if, and only if, $\vec{x}$ can be written as a linear combination of $\vec{v}_1$ and $\vec{v}_1 + \vec{v}_2$.